

IMO(50th) – 2009

First Day

1. Let n be a positive integer and let $a_1, \dots, a_k$ ($k \geq 2$) be distinct integers in the set $\{1, 2, \dots, n\}$ such that n divides $a_i(a_{i+1} - 1)$ for $i = 1, 2, \dots, k-1$. Prove that n does not divide $a_k(a_1 - 1)$.
2. Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K , L and M be the midpoints of the segments BP , CQ and PQ respectively, and let Γ be the circle passing through K , L and M . Suppose the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.
3. Suppose that $s_1, s_2, s_3, \dots$ is a strictly increasing sequence of positive integers such that the subsequences $s_1, s_2, s_3, \dots$ and $s_{s_1+1}, s_{s_2+1}, s_{s_3+1}, \dots$ are both arithmetic progressions. Prove that the sequence $s_1, s_2, s_3, \dots$ is itself an arithmetic progression.

Second Day

4. Let ABC be a triangle with $AB = AC$. The angle bisector of $\angle CAB$ and $\angle ABC$ meet the sides BC and CA at D and E respectively. Let K be the incentre of triangle ADC . Suppose that $\angle BEK = 45^\circ$. Find all possible values of $\angle CAB$.
5. Determine all functions f from the set of positive integers to the set of positive integers such that, for all positive integers a and b , there exist a non-degenerate triangle with sides of lengths a , $f(b)$, $f(b + f(a) - 1)$. (A triangle is non-degenerate if its vertices are not collinear.)
6. Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n-1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen in such a way that the grasshopper never lands on any point in M .
